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ITA0611 – Machine Learning

IET Assignment

Design and Development of an Intelligent Early-Warning System for Crop Disease and Yield Risk Prediction Using Machine Learning

Department Computer Science and Engineering

Programme Artificial Intelligence & Data Science

Course Code & Course

Name

ITA0611 – Machine Learning

Academic Year / Batch 2024

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Assignment Title Design and Development of an Intelligent Early-Warning System for Crop Disease and Yield Risk Prediction Using Machine Learning

Date of Issue 09-02-2026

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Maximum Marks 100

Course Outcome(s) – CO CO6 – Instance-based learning (KNN, LWR, case-based reasoning); CO7

– Data manipulation/analysis with NumPy & Pandas; CO8 – Integration

of theoretical and practical ML knowledge

Bloom's Taxonomy Level L5 – Evaluate; L6 – Create

SDG Mapping SDG 2 – Zero Hunger; SDG 9 – Industry, Innovation & Infrastructure;

SDG 13 – Climate Action

Industry / Societal

Relevance

Precision-agriculture decision support; early crop-disease alerting for smallholder and commercial farmers

1. Problem Understanding and Formulation

What is the problem? Crop losses caused by disease outbreaks and adverse weather/soil conditions are difficult

for farmers to anticipate because the interaction between environmental, soil and crop variables is complex and

non-linear. The assignment requires an intelligent system that ingests environmental, soil, crop and

historical-yield data and predicts (a) the disease-risk category of a field and (b) the expected yield, so that preventive action can be taken before losses occur.

Expected outcomes: a cleaned and analysed dataset; a concept-learning formulation with a worked Candidate-Elimination example; a decision tree that produces interpretable disease-risk rules; a Bayesian classifier calibrated by Maximum Likelihood Estimation; K-Nearest-Neighbour and Locally Weighted Regression

models for classification and yield regression respectively; a backpropagation-trained neural network; a genetic-algorithm-based feature-selection routine; a comparative evaluation across all models; and a single

integrated function that turns a new sensor/field reading into a risk label, a yield estimate and a plain-language

recommendation.

Data available: no ready-made field dataset was supplied with the problem statement, so a simulation-based

dataset of 1,200 cleaned records was constructed in Python (NumPy/Pandas) with agronomically grounded relationships — e.g. fungal disease pressure rising with humidity and rainfall, and yield falling with heat stress and disease severity while rising with soil moisture and nitrogen. This keeps every subsequent analysis fully reproducible (fixed random seed) and free of copied third-party data.

Assumptions: (1) the 15-day rolling rainfall/humidity/temperature window is representative of the disease-conducive micro-climate; (2) disease symptom severity is observable through a 0–100 sensor/visual

index; (3) soil nutrient levels (N, P, K) and pH are measured periodically and are approximately stable within a short prediction horizon; (4) the five crops modelled (rice, wheat, maize, cotton, sugarcane) share comparable

disease-risk drivers for the purposes of this assignment.

Constraints: the models must be interpretable enough for a non-expert farmer/extension officer to act on (favouring decision trees and rule-based summaries alongside the neural network); the system must run on modest hardware (no GPU dependency); and every learning method specified in the problem statement (concept learning, decision tree, Bayesian, KNN, LWR, MLP+backpropagation, GA) had to be implemented and compared rather than only the best-performing one used.

2. Application of Course Knowledge

The project draws directly on CO6–CO8 of the course. Concept learning (Mitchell's Candidate-Elimination algorithm) is applied to formalise disease-risk detection as a search through a hypothesis space bounded by a

most-general (G) and most-specific (S) boundary set. Decision-tree learning uses information-theoretic attribute

selection (entropy / information gain, and Gini impurity) to recursively partition the feature space. Bayesian learning applies Bayes' theorem together with Maximum Likelihood Estimation of class priors $P(\text{class})$ and per-class Gaussian likelihood parameters (mean, variance) to obtain a generative classifier. Instance-based learning is represented by K-Nearest-Neighbours (a lazy, non-parametric classifier using Euclidean distance) and

Locally Weighted Regression (a kernel-weighted local linear model solved via the weighted normal equations $\theta =$

$(X^T W X)^{-1} X^T W y$). A multilayer perceptron is trained with manually derived forward and backward propagation equations (ReLU hidden layer, softmax output, cross-entropy loss, gradient descent). A genetic algorithm — encoding feature subsets as binary chromosomes, using tournament selection, single-point crossover, bit-flip

mutation and elitism — searches for a compact, high-performing feature subset. All numerical work (cleaning,

aggregation, feature engineering, evaluation metrics) uses NumPy and Pandas as required by CO7.

3. Solution / Design / Methodology

The methodology follows the ten questions of the problem statement in sequence. Each stage is implemented as

an independent, reproducible Python module; outputs from earlier stages (the cleaned dataset, the trained

decision tree, the GA-selected features) feed into later stages, culminating in the integrated system of

Question

10. Question 1 — Data Collection and Preprocessing

A 1,208-record simulated dataset (five crops; temperature, rainfall, humidity, soil moisture, soil pH, N-P-K, previous yield, a disease symptom index, disease-risk label and continuous yield) was generated with NumPy and

cleaned with Pandas: 8 duplicate rows were dropped, missing soil-pH values (25 records) were imputed with the

median and missing humidity values (15 records) with the mean, categorical crop type was one-hot encoded,

and all numeric features were z-score standardised for the distance- and gradient-based models. The cleaned

dataset has 1,200 records.

Figure 1. Class distribution of the disease-risk target after cleaning (Medium is the modal class, reflecting realistic field conditions).

Figure 2. Correlation matrix of environmental, soil and yield attributes — humidity and rainfall show the strongest positive correlation with

the symptom score, motivating their later selection by both the decision tree and the genetic algorithm.

Question 2 — Concept Learning and Version Space

The target concept was formulated as $\text{HighDiseaseRisk}(\text{Temperature}, \text{Humidity}, \text{Rainfall}, \text{SoilMoisture}) \rightarrow \{\text{Yes}, \text{No}\}$, with each attribute discretised into $\{\text{Low}, \text{Moderate}, \text{High}\}$. The hypothesis space is the set of conjunctions

over these four attributes, each position taking a specific value or '?' (don't-care); the inductive bias of Candidate-Elimination is therefore the conjunctive-hypothesis bias — it can only represent concepts expressible

as a single conjunction of attribute values, and it assumes the target concept is contained in this hypothesis space.

A worked six-example trace (implemented from first principles, not a library

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